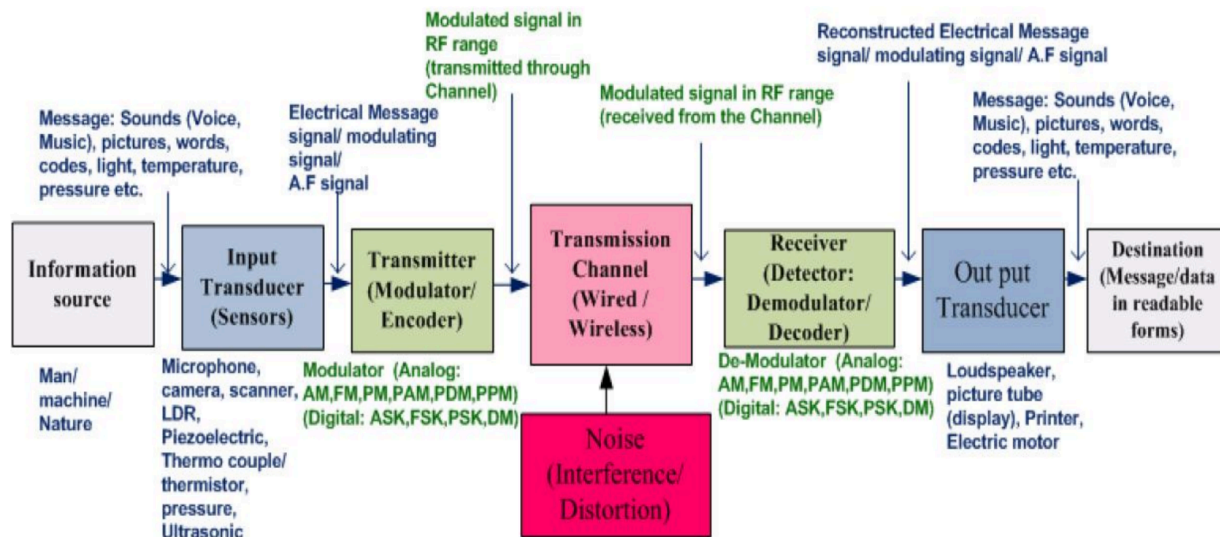


**Q1. Draw and explain the basic block diagram of a communication system.**

A communication system is a system that transfers information from a source to a destination through a communication channel. It consists of various functional blocks that process, transmit, receive, and reproduce the original information.



**Q2. Give examples of communication systems used in day-to-day life. (2 Marks)**

Examples of communication systems include:

1. Mobile phone communication
2. Television broadcasting
3. Radio broadcasting
4. Wi-Fi communication
5. Bluetooth communication
6. Satellite communication
7. Internet communication
8. Optical fibre communication

### Q3. Explain the Need of Modulation. (5 Marks)

Modulation is the process of varying a characteristic of a high-frequency carrier wave according to the instantaneous value of the low-frequency message signal. The resulting wave is called the modulated wave or radio wave.

#### Modulation Need

- ① Modulating a signal helps reduce the height of antenna required to transmit it  
Height of Antenna  $\propto \lambda/2$
- ② Modulation of Signal increases its power making it possible to transmit over long distance  
Power radiated  $\propto \frac{1}{\lambda^2}$
- ③ Radiation of electrical energy is only possible at frequency  $> 20\text{ kHz}$ .  
 $\therefore$  Modulation makes it possible to transmit without wires
- ④ Multiplexing  
multiple signal with frequencies in close range cannot travel through same channel.  
modulating helps by changing the frequency to ensure they are not in close range
- ⑤ Equipments work on a certain range of frequencies

**Q4. Explain how modulation reduces antenna height with an example.**

I. Reduction in Antenna Height ( $\lambda/2$ )

$$\text{wavelength} = \frac{\text{Velocity}}{\text{frequency}} = \frac{3 \times 10^8}{\text{frequency in (Hz)}} \text{ meter}$$

) Take frequency – 10 KHz and calculate antenna height  $\lambda = 30 \text{ km}$   $\lambda/2 = 15 \text{ km}$

) Take frequency – 100 MHz and calculate antenna height  $\lambda = 3 \text{ m}$   $\lambda/2 = 1.5 \text{ m}$

Explain the equation in the above image

**Q5. Define Signal-to-Noise Ratio (SNR), Noise Factor (F), and Noise Figure (NF).**

### 1. Signal-to-Noise Ratio (SNR)

SNR is the ratio of signal power to noise power.

$$SNR = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad SNR_{\text{dB}} = 10 \log_{10} \left( \frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

### 2. Noise Factor (F)

Noise Factor is the ratio of input SNR to output SNR.

$$F = \frac{(S/N)_{\text{input}}}{(S/N)_{\text{output}}}$$

Ideal value = 1

### 3. Noise Figure (NF)

Noise Figure is the Noise Factor expressed in decibels.

$$NF = 10 \log_{10}(F)$$

**Q6. Explain the significance of SNR, Noise Factor, and Noise Figure. (5 Marks)**

**Significance of SNR**

- Measures signal quality.
- Higher SNR → Better communication.
- Lower SNR → More noise.

According to the lecture notes:

- **SNR < 0 dB** → Poor communication
- **10–20 dB** → Suitable for voice/data
- **>30 dB** → Near-perfect communication

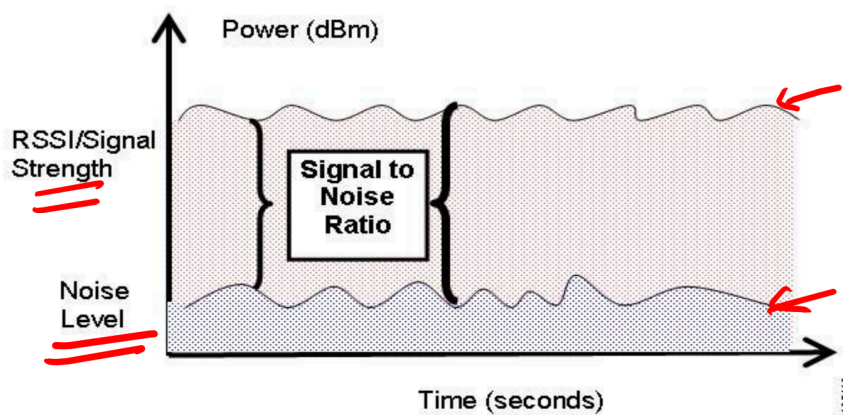
**Significance of Noise Factor**

- Indicates how much the system degrades the input SNR.
- Lower Noise Factor is desirable.
- Ideal value is 1.

**Significance of Noise Figure**

- Noise Figure expresses Noise Factor in dB.
- Lower Noise Figure means better system performance.
- Ideal value is **0 dB**.

**Q7. With only a diagram explain Signal-to-Noise Ratio**



### Q9. Define Amplitude Modulation (AM). (2 Marks)

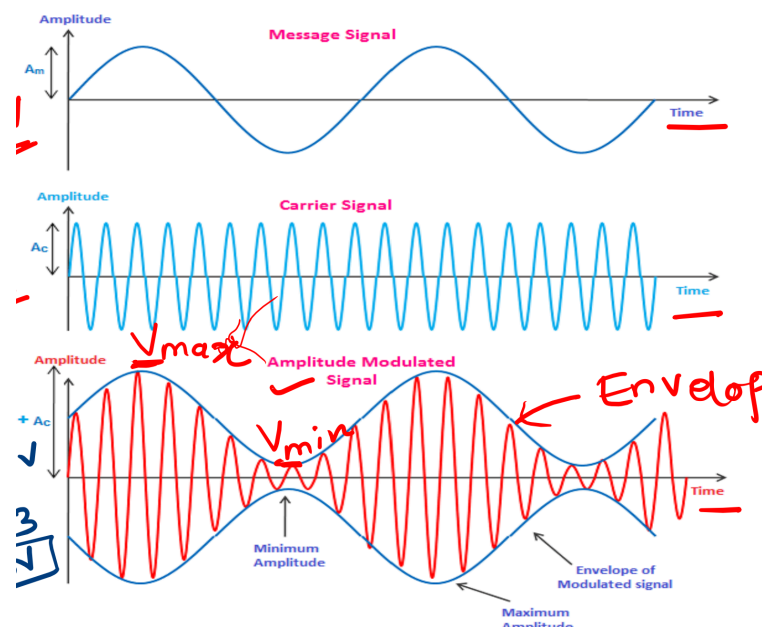
#### Definition

Amplitude Modulation (AM) is the process of varying the amplitude of a high-frequency carrier wave according to the instantaneous value of the message signal, while keeping the frequency and phase of the carrier constant. The resulting signal is called the AM wave.

#### Characteristics

- Carrier amplitude varies.
- Carrier frequency remains constant.
- The carrier phase remains constant.
- Used in commercial AM radio broadcasting and CB radio communication.

### Q10. Draw and explain the Time Domain and Frequency Domain representation of an AM Wave.



#### Explanation

- The amplitude of the carrier changes according to the message signal.
- The envelope of the AM wave has the same shape as the message signal.
- The frequency and phase of the carrier remain unchanged.

### Q11. Derive the mathematical expression for an AM Wave

## Derivation of AM

Let modulating signal be

  $V_m = V_m \sin \omega_m t$  — (1)  
 $\omega_m = 2\pi f_m$

Let carrier signal be

  $V_c = V_c \sin \omega_c t$  — (2)  $f_c > f_m$

Let amplitude of AM wave is

$V_{am} = V_c + V_m$  ... by definition  
Substituting eqn (1)

$$V_{am} = V_c + V_m \sin \omega_m t$$
$$= V_c \left[ 1 + \frac{V_m \sin \omega_m t}{V_c} \right]$$

Modulation index ( $m$ ) =  $\frac{V_m}{V_c}$   
[A unitless ratio b/w 0 & 1]

It signifies the amount of change in modulated signal

$$V_{am} = V_c [1 + m \sin \omega_m t] \text{ — (3)}$$

$$V_{am}(t) = V_{am} \sin \omega_c t$$

$$V_{am}(t) = V_c [1 + m \sin \omega_m t] \sin \omega_c t$$

$$V_{am}(t) = V_c \sin \omega_c t + \underbrace{m V_c \sin \omega_c t \sin \omega_m t}_{\sin A \sin B}$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$V_{am}(t) = \underbrace{V_c \sin \omega_c t}_{\text{Carrier Signal}} + \frac{m V_c}{2} \left[ \underbrace{\cos(\omega_c - \omega_m)t}_{\text{Lower Side band}} - \underbrace{\cos(\omega_c + \omega_m)t}_{\text{Upper Side band}} \right]$$

**Q13. Explain Modulation Index with examples. (5 Marks)**

The Modulation Index is the ratio of the amplitude of the modulating signal to the amplitude of the carrier signal.

$$m = \frac{V_m}{V_c}$$
$$m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

| Modulation Index | Type                       |
|------------------|----------------------------|
| $m < 1$          | Under Modulation           |
| $m = 1$          | Critical (100%) Modulation |
| $m > 1$          | Over Modulation            |

**Examples (From Lecture Notes)**

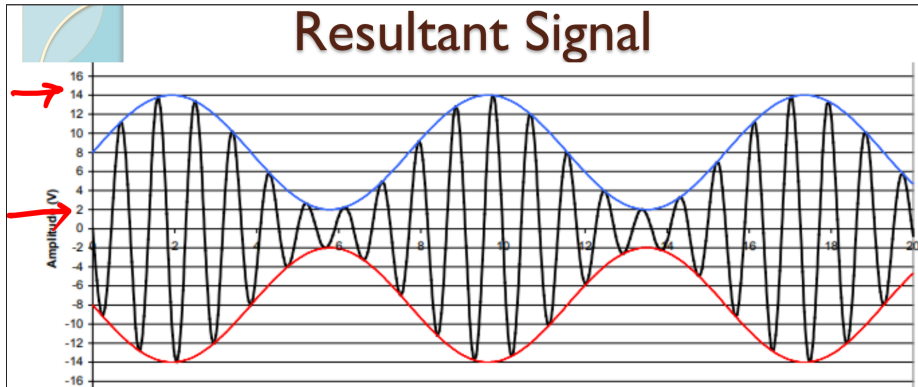
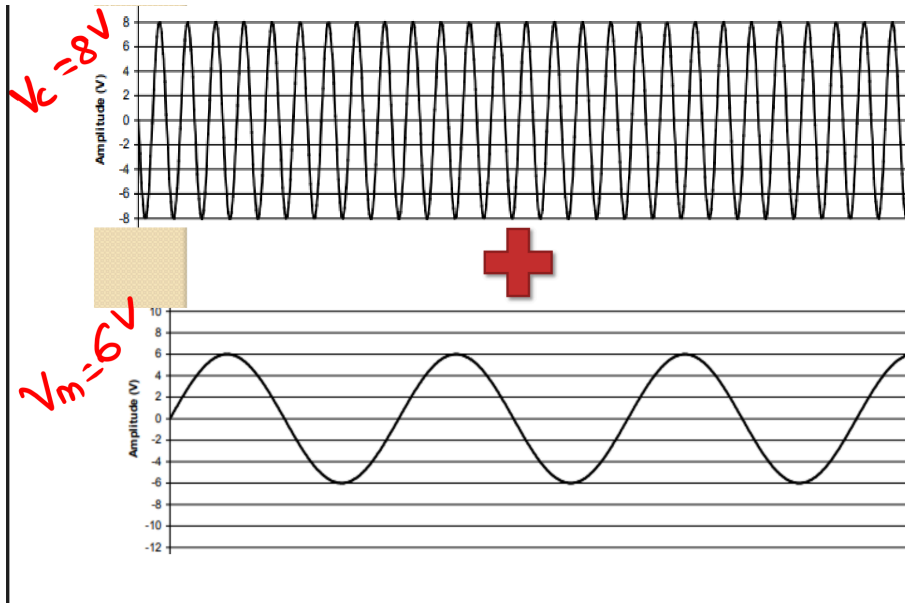
Carrier Voltage = 8 V

| Message Voltage | Modulation Index | Percentage |
|-----------------|------------------|------------|
| 2 V             | 0.25             | 25%        |
| 4 V             | 0.50             | 50%        |
| 6 V             | 0.75             | 75%        |
| 8 V             | 1.00             | 100%       |
| 10 V            | 1.25             | 125%       |

**Q14. Explain Under Modulation, Critical Modulation and Over Modulation with waveforms. (5 Marks)**

### 1. Under Modulation

**Condition:**  $m < 1$



$$V_{\max} = V_c + V_m = 8 + 6 = 14 \text{ V}$$

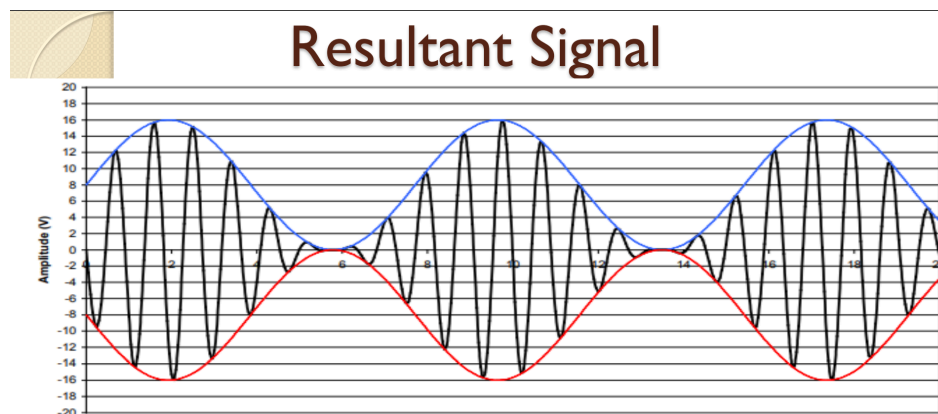
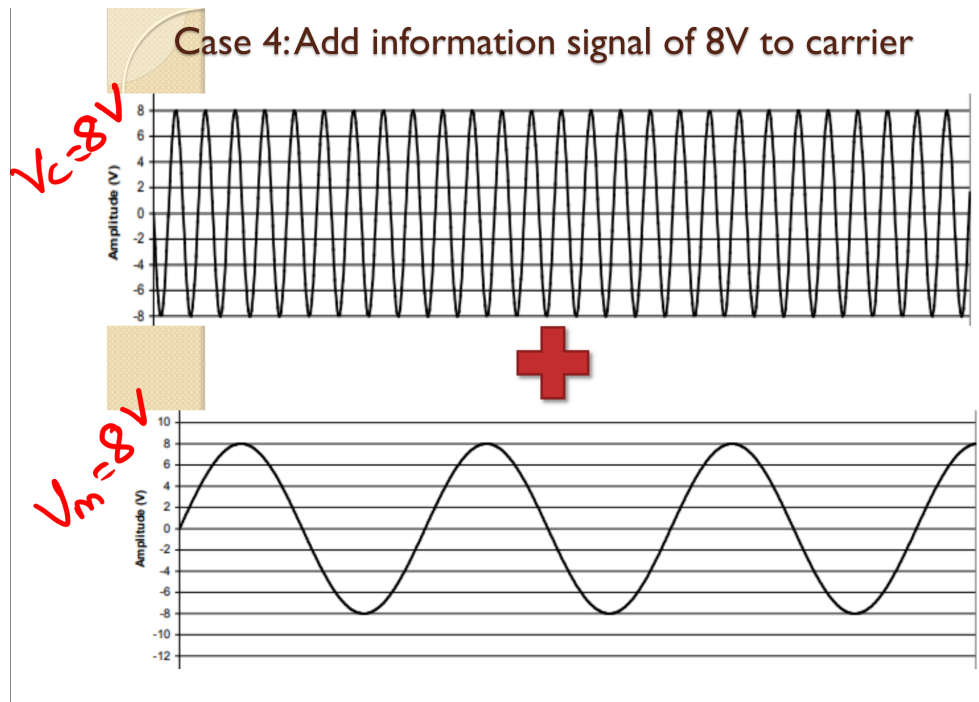
$$V_{\min} = V_c - V_m = 8 - 6 = 2 \text{ V}$$

$$m = \frac{12}{16} \times 100\% = 75\%$$



## 2. Critical (100%) Modulation

Condition:  $m=1$



$$V_{\max} = V_c + V_m = 8 + 8 = 16V$$

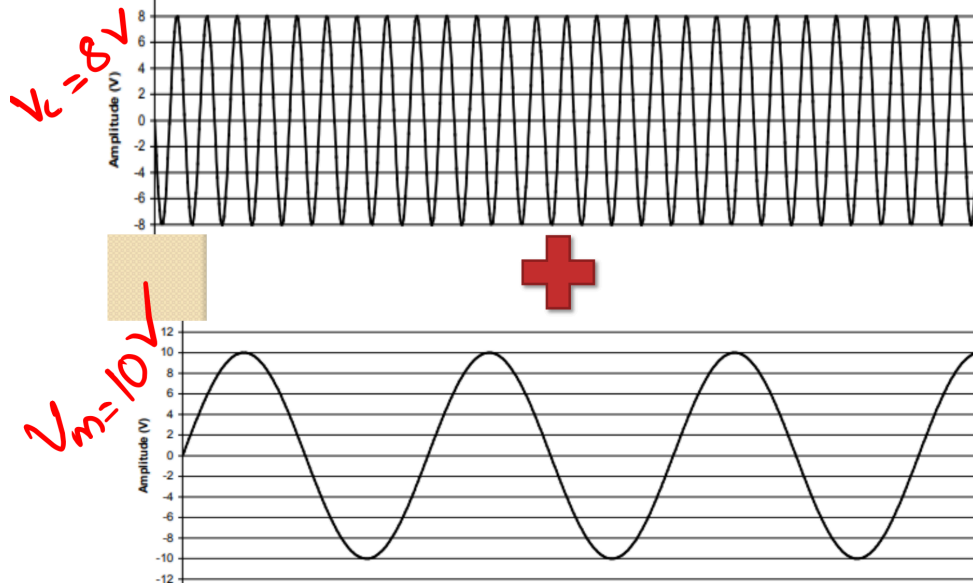
$$V_{\min} = V_c - V_m = 8 - 8 = 0V$$

$$m = \frac{16}{16} \times 100\% = 100\%$$

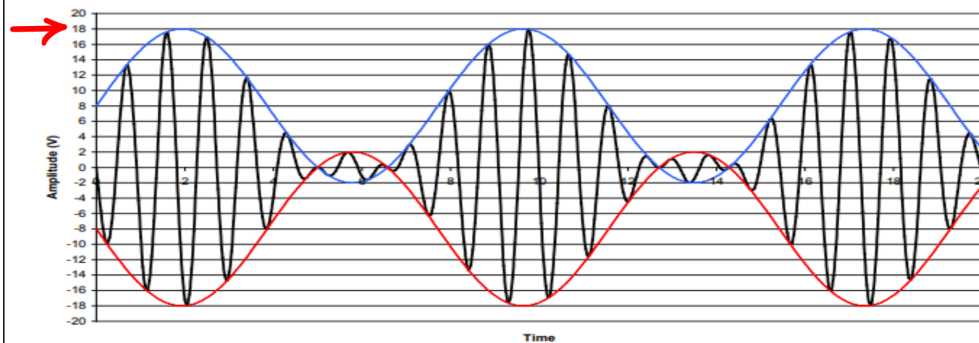
### 3. Over Modulation

Condition:  $m > 1$

Case 5: Add information signal of 10V to carrier



### Resultant Signal



*Over Modulation*

$$V_{\max} = V_c + V_m = 8 + 10 = 18V$$
$$V_{\min} = V_c - V_m = 8 - 10 = \underline{\underline{-2V}}$$

$$m = \frac{20}{16} \times 100\% = 125\%$$